GENERIC EXACT RECONSTRUCTION OF REFLECTED-CAP DATA FROM AN ERASED FLAT BOUNDARY

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ABSTRACT. Let a convex three-polytope in a hyperplane of $\mathbb{R}^4$ carry one reflected pyramidal cap on each facet, with the caps independently rotated about their facets. We study the proper-zero regime: at least one cap remains nonflat and at least one cap returns to the core hyperplane. The flat caps then merge with the core and erase their internal source-layer boundaries.

We define an intrinsic boundary-germ skeleton of the merged flat component. Its vertices satisfying a perpendicular-bisector neighbour test are exactly the hidden reflected apexes, provided no core vertex satisfies the same test. This yields exact reconstruction and uniqueness of every compatible decomposition. For each fixed realized core and visible-facet mask, centres violating the hypothesis lie in a finite union of proper spheres and quadrics, and hence form a measure-zero set. We also give an exact rational reference algorithm and report a complete 16,744-instance finite replay. The finite replay supports the implementation; it is not used as the proof.

1. Introduction

Inverse Voronoi and Dirichlet problems usually begin with a supplied tessellation, cell complex, embedded diagram, or collection of cell boundaries. The input considered here is coarser. Several reflected sites and their facet boundaries have been erased by a coplanar union before the inverse problem begins.

The model is elementary. A convex polytope P lies in a three-dimensional hyperplane $H \subset \mathbb{R}^4$, an interior point o is fixed, and the reflection of o in each facet plane is used as the apex of a pyramid. Each pyramid is then rotated through an independent angle about its base facet. When an angle is zero, the corresponding pyramid is coplanar with P. All such flat pyramids merge with P into one three-dimensional subset $M_J \subset H$. Only the raw union is observed; cell labels and internal flat seams are not.

The central issue is therefore not the forward reflection construction. That construction is classical and appears explicitly in Ash and Bolker's Theorem 2 [2]. The issue is whether the hidden reflected sites can be identified from the exposed boundary after the internal seams have disappeared.

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A naive graph is insufficient. Taking the polygonal one-skeleton after every coplanar merge can swallow a collinear point at which the local boundary germ changes. Retaining the sides of the original pyramids repairs that example but illegitimately remembers the unknown decomposition. We instead use the noncoplanar crease graph of the maximal exposed planar sheet germs. It keeps a collinear point exactly when the local incident-sheet germ changes, and removes it when it is only a presentation subdivision.

Our main result is conditional but size-uniform.

Theorem 1.1 (informal)**.** *In dimension three, suppose a reflected-cap datum is proper-zero and no core vertex passes the intrinsic bisector-neighbour test. Then the passing vertices are precisely the hidden reflected apexes, the core is recovered by their bisector halfspaces together with the visible hinge halfspaces, and the datum is unique up to ambient Euclidean congruence. For a fixed realized core and visible mask, the excluded centres form a measure-zero subset of the core.*

The theorem does not assert uniqueness at excluded centres, a result in higher dimension, or novelty or priority relative to all possible prior art. Those boundaries are repeated in Section 9.

2. The reflected-cap model and exact contacts

Let $H \cong \mathbb{R}^3$ be a fixed hyperplane in $\mathbb{R}^4$, and let e be a unit normal to H. Let $P \subset H$ be a full-dimensional convex polytope and $o \in \operatorname{int} P$. Translate o to the origin and write the irredundant unit-normal description

$$P = \{x \in H : n_i \cdot x \le h_i,\ i = 1, \dots, m\}, \qquad \|n_i\| = 1, \quad h_i > 0,$$

with facets $F_i = P \cap \{n_i \cdot x = h_i\}$. For $\theta_i \in (-\pi, \pi)$ set

$$a_i(\theta_i) = h_i(1 + \cos\theta_i)n_i + h_i \sin\theta_i e, \qquad C_i(\theta_i) = \operatorname{conv}(F_i \cup \{a_i(\theta_i)\}).$$

The excluded value π collapses the apex to o. At $\theta_i = 0$, $a_i = 2h_i n_i$, the reflection of o in $\operatorname{aff} F_i$.

Two data are called congruent when an ambient Euclidean isometry carries the core, centre, and unordered cap family of one to those of the other. Thus the angle coordinates are used only to parametrise the geometric caps.

The datum is $D = (P, o, \theta)$ and its raw union is

$$X(D) = P \cup \bigcup_{i=1}^{m} C_i(\theta_i) \subset \mathbb{R}^4,$$

retained as a bare Euclidean subset. Put

$$J = \{i : \theta_i = 0\}, \qquad I = [m] \setminus J.$$

The datum is *proper-zero* when I and J are both nonempty. Its merged flat component is

$$M_J = P \cup \bigcup_{j \in J} C_j(0) \subset H.$$

Proposition 2.1 (exact contacts)**.** *For every angle vector $\theta \in (-\pi, \pi)^m$,*

$$P \cap C_i(\theta_i) = F_i, \qquad C_i(\theta_i) \cap C_j(\theta_j) = F_i \cap F_j \quad (i \neq j).$$

Thus the cells form an embedded face-to-face polyhedral complex whose contact poset is independent of the angles.

Proof. Use the Voronoi diagram of the sites $0, a_1(\theta_1), \ldots, a_m(\theta_m)$. The sites are distinct: equality of two nonzero H-components would force two irredundant outward facet normals to agree. For $x \in P$,

$$\begin{aligned}\|x - a_i\|^2 - \|x\|^2 &= \|a_i\|^2 - 2x \cdot a_i \\ &= 2h_i(1 + \cos\theta_i)(h_i - n_i \cdot x) \geq 0.\end{aligned}$$

The first factor is positive on $(-\pi, \pi)$. Hence P lies in the Voronoi cell V_0 of 0, and $F_i \subset V_0 \cap V_i$. The apex a_i is in the interior of V_i. Convexity of V_i gives $C_i \setminus F_i \subset \operatorname{int} V_i$. Therefore no point of $P \cap C_i$ can lie outside F_i.

If a point of $C_i \cap C_j$ lay outside one of the two hinge facets, the same argument would place it in one open Voronoi cell and simultaneously in a different open cell or in V_0. This is impossible. The reverse inclusion $F_i \cap F_j \subset C_i \cap C_j$ is immediate. □

3. Intrinsic recovery in the proper-zero regime

For a three-dimensional polyhedral subset $Y \subset \mathbb{R}^4$, let $\mathrm{FlatReg}_3(Y)$ be the set of points having a neighbourhood in Y that is relatively open in one affine three-plane. This set is determined by Y alone.

Proposition 3.1 (proper-zero intrinsic recovery)**.** *If $I \neq \varnothing$, the raw subset $X(D)$ intrinsically determines M_J, every nonflat cap and its hinge facet, the centre o, and the fully reflected site associated with every nonflat hinge.*

Proof. The closures of the connected components of $\mathrm{FlatReg}_3(X)$ are M_J and the nonflat caps. Indeed, the zero-angle cells join the core through facets inside H and form one three-dimensional flat component, while every nonflat cap has a different affine hull. Proposition 2.1 accounts for every contact. The codimension-one adjacency graph of these closures is a star.

If $|I| \geq 2$, the unique star centre is M_J. If $|I| = 1$, let i be the nonflat index, let $V = \mathrm{vol}_3(P)$, and let $v_k = \mathrm{vol}_3 \operatorname{conv}(\{o\} \cup F_k)$. The radial pyramids tile P, so $\sum_k v_k = V$. Rotation preserves the intrinsic volume of a cap, and hence

$$\mathrm{vol}_3(M_J) = 2V - v_i > v_i = \mathrm{vol}_3(C_i).$$

Thus in the two-component case the larger component is M_J.

Fix one recovered nonflat cap, with apex A_i and hinge F_i. Let p_i be the orthogonal projection of A_i to $\operatorname{aff} F_i$, let $r_i = \operatorname{dist}(A_i, \operatorname{aff} F_i)$, and let ν_i be either unit normal to $\operatorname{aff} F_i$ inside H. Rotation about $\operatorname{aff} F_i$ preserves the radius in its two-dimensional normal plane, so

$$\{p_i - r_i\nu_i, p_i + r_i\nu_i\} = \{o, a_i(0)\}.$$

The centre belongs to the interior of M_J. The other point does not belong to M_J: apply Proposition 2.1 to the angle vector in which cap i is also set to zero. Its apex meets neither the core nor another cap away from prescribed lower-dimensional contacts. Consequently o is the unique one of the two candidates in $\operatorname{int} M_J$. Reflection of o in the recovered hinge then gives the fully reflected site. □

Every compatible decomposition of a given proper-zero raw union therefore has the same M_J, centre, nonflat caps, and visible hinge facets.

4. The intrinsic boundary-germ skeleton

Let an *exposed planar sheet* of ∂M_J be the closure of a maximal connected planar piece of its relative-regular locus. Internal coplanar subdivisions are not retained.

Definition 4.1 (intrinsic germ skeleton)**.** Atomise the boundaries of all exposed sheets at every endpoint contributed by any other sheet. Retain the open segments carried by at least two distinct supporting planes. Suppress a collinear degree-two point only if its complete local incident-sheet germ is unchanged through that point. The resulting canonical graph is $\Gamma(M_J)$. A vertex a *passes* if every neighbour of a in $\Gamma(M_J)$ belongs to the perpendicular bisector

$$B(o, a) = \{x \in H : \|x - o\| = \|x - a\|\}.$$

The definition is intrinsic: exposed sheet subsets, their supporting planes, and their local germs are properties of M_J as a point set. In particular, the graph stores neither a hidden cap label nor an erased internal seam.

A compatible decomposition of M_J consists of a convex core P', the already recovered visible hinge facets, and hidden facets G_a indexed by reflected sites a such that

$$M_J = P' \cup \bigcup_{a \in A} \operatorname{conv}(G_a \cup \{a\})$$

with pairwise disjoint cell interiors. For an edge f of G_a, write $W(a, f) = \operatorname{conv}(\{a\} \cup f)$ for the corresponding wall.

Lemma 4.2 (boundary inventory)**.** *For every compatible decomposition,*

$$\partial M_J = \bigcup_{i \in I} F_i \ \cup \ \bigcup_{a \in A} \ \bigcup_{f \in E(G_a)} W(a, f).$$

Proof. The boundary of P' is the union of its visible facets and hidden facets. Each hidden facet is shared with its cap and becomes internal to M_J. The remaining cap faces are the walls. Proposition 2.1 says that a wall meets another cell only in a lower-dimensional face. □

Lemma 4.3 (decomposition computation of the intrinsic graph)**.** *Starting from the inventory in Lemma 4.2, erase a side across which the two incident boundary patches are coplanar, atomise at every other sheet-boundary* *endpoint, and suppress only unchanged collinear degree-two germs. The result is $\Gamma(M_J)$, independently of the compatible decomposition used for the computation.*

Proof. At a relative-interior point of a patch side, exact face-to-face contact gives no transverse crossing in a patch interior. If the two incident patch planes agree, their union is locally planar and the side is absent from the crease locus. If the planes differ, the side is an intrinsic noncoplanar crease. Passing to maximal planar sheets deletes exactly the first kind. Atomisation restores every endpoint where another sheet germ begins, ends, or changes, including a collinear endpoint omitted by a polygon hull. The final suppression removes only a presentation subdivision at which the complete local germ is unchanged. These conditions depend only on ∂M_J. □

Remark 4.4 (the swallowed-corner control)**.** There are legitimate data for which two coplanar merged triangles have a collinear boundary point that a polygonal hull would drop. At that point a noncoplanar incident sheet changes, so Definition 4.1 retains it. The exact control used in the artifact is

$$P = \{x \geq 0,\ z \geq 0,\ x + y \geq 2,\ y \leq 6,\ x + z \leq 4\}, \qquad o = (1, 2, 1),$$

with the hidden facets $x = 0$ and $z = 0$. Their shared edge starts at the foot of the perpendicular from o, making the two apexes and that endpoint collinear.

5. The apex signature and uniqueness

Lemma 5.1 (only its own walls contain an apex)**.** *For $a \in A$, the boundary patches containing a are exactly the walls of the cap with apex a.*

Proof. The reflection a lies strictly outside P'. For $b \neq a$, Proposition 2.1 gives $C_a \cap C_b = G_a \cap G_b \subset P'$, so a belongs to no wall of C_b. Lemma 4.2 leaves no other boundary patch. □

Lemma 5.2 (candidate types)**.** *Every vertex of $\Gamma(M_J)$ is either a hidden apex or a vertex of the compatible core.*

Proof. By Lemma 4.3, a graph vertex is a branch point or a retained endpoint of the patch-side arrangement. Exact face-to-face contacts create no transverse crossing in a patch interior, so such a point is a patch corner. Visible-facet corners are core vertices, and wall corners are one apex and two core vertices. □

Lemma 5.3 (apexes pass)**.** *Every hidden apex passes in $\Gamma(M_J)$.*

Proof. Let a be the apex over a hidden facet G_a. For each vertex v of G_a, the two walls over the two facet edges incident to v meet along $[a, v]$. Their planes are distinct: if they were coplanar, the apex and three consecutive vertices of the convex facet polygon would be coplanar, forcing $a \in \operatorname{aff} G_a$, contrary to $o \in \operatorname{int} P'$. Thus the relative interior of $[a, v]$ is an intrinsic crease. The apex is a branch point, and v is retained because the two wall-sheet germs containing a end or change there. Conversely, Lemma 5.1 shows that every crease incident to a is of this form. Hence

$$N_{\Gamma(M_J)}(a) = \operatorname{vert} G_a \subset \operatorname{aff} G_a = B(o,a).$$

□

For a facet f of a compatible core, write $v_f = \operatorname{vol}_3 \operatorname{conv}(\{o\} \cup f)$.

Lemma 5.4 (volume identity)**.** *Let $W = \sum_{i\in I} v_{F_i}$. Every compatible decomposition satisfies*

$$\operatorname{vol}_3(M_J) = W + 2\sum_{a\in A} v_{G_a}.$$

Consequently all compatible cores have the same volume.

Proof. The radial pyramids from o to the facets tile a convex core, so $\operatorname{vol}_3(P') = W + \sum_a v_{G_a}$. The cap over G_a is a reflection of its radial pyramid and has the same volume. Exact contacts give pairwise disjoint cell interiors, proving the displayed identity. Both M_J and W are intrinsic, so $\sum_a v_{G_a}$ and then the core volume are common to all decompositions. □

Lemma 5.5 (nested site sets)**.** *If (P', A) and (P'', A'') are compatible and $A'' \subseteq A$, then the two decompositions coincide.*

Proof. The hidden facet associated with a has halfspace $\{x : \|x - o\| \leq \|x - a\|\}$. Therefore

$$P' = \bigcap_{i\in I} H_i^- \cap \bigcap_{a\in A} \{x : \|x - o\| \leq \|x - a\|\},$$

and similarly for P''. The inclusion $A'' \subseteq A$ gives $P' \subseteq P''$. Lemma 5.4 gives equal finite volume, so the two closed convex bodies agree. Their active facets and reflected sites then agree. □

Theorem 5.6 (exact reconstruction)**.** *Let D be a three-dimensional proper-zero datum. Suppose no vertex of its core passes in $\Gamma(M_J)$. Then the passing vertices of $\Gamma(M_J)$ are exactly the hidden reflected apexes, the core is*

$$P = \bigcap_{i\in I} H_i^- \cap \bigcap_{a \text{ passing}} \{x : \|x - o\| \leq \|x - a\|\},$$

and every compatible decomposition of $X(D)$ is the same. Consequently, $X(D)$ congruent to $X(D')$ implies D congruent to D'.

Proof. Let A be the hidden apex set of D, and let (P'', A'') be any compatible decomposition. The graph $\Gamma(M_J)$ is intrinsic, hence common to both decompositions. Lemma 5.3 gives that every element of A and A'' passes in this common graph. By Lemma 5.2, a passing vertex is an apex of D or a core vertex. The second possibility is excluded by hypothesis, so the passing set is exactly A and $A'' \subseteq A$. Lemma 5.5 completes the uniqueness proof.

An ambient isometry carrying one raw union to another preserves the intrinsic components, centre, exposed sheet germs, and $\Gamma(M_J)$. It carries the first datum to a compatible decomposition of the second raw union, which uniqueness identifies with the second datum. □

6. The exceptional centre set

Theorem 6.1 (fixed-realization genericity)**.** *Fix a realized three-polytope P and a visible-facet mask. The set of centres $o \in \operatorname{int} P$ at which some core vertex passes is contained in a finite union of proper spheres and quadrics. In particular, it has three-dimensional Lebesgue measure zero.*

Proof. Use fixed absolute coordinates. Suppose a core vertex v is incident to a hidden facet with outward unit normal n and equation $n \cdot x = h$. The reflected apex is

$$a = o + 2(h - n \cdot o)n.$$

The two walls meeting along $[v, a]$ have distinct planes, so this is an edge of $\Gamma(M_J)$. If v passes, then $a \in B(o, v)$, which forces

$$4(h - n \cdot o)^2 = \|v - o\|^2.$$

Its quadratic part is $o^{\mathsf{T}}(4nn^{\mathsf{T}} - I)o$. The eigenvalue along n is 3 and the two eigenvalues on $n^{\perp}$ are -1. The polynomial is nonzero and its zero set is a proper quadric.

If v is incident only to visible hinge facets, the core edges at v are intersections of distinct facet planes and are intrinsic crease edges. Any crease neighbour w required by passing satisfies

$$\|w - o\|^2 = \|w - v\|^2,$$

a proper sphere condition in o.

There are finitely many vertices and incidences. Every passing event is contained in at least one of the displayed proper algebraic hypersurfaces, and a finite union of such zero sets has measure zero. □

The theorem deliberately does not assert that the bad-centre set itself is closed. A subset of a closed algebraic union need not be closed. Nor does the argument define a measure on simultaneously varying realizations of P.

7. Exact reconstruction algorithm and evidence

Assume an exact rational polygonal presentation of the exposed boundary. Coplanar subdivisions are presentation data and may be inserted or removed. The conceptual algorithm is:

1. Compute the closures of the connected components of the local-flat locus. With at least three components, select the unique centre of their codimension-one adjacency star; with exactly two, select the component of larger intrinsic volume. This is M_J.
2. Intersect every other component with M_J to recover the nonflat hinges and their cap geometry.
3. Recover o from either nonflat cap by the two-candidate construction in Proposition 3.1.

4. Construct $\Gamma(M_J)$ from exposed sheet germs.
5. Keep exactly the graph vertices that pass.
6. Intersect the visible hinge halfspaces with the bisector halfspaces of the passing vertices.

Under the hypothesis of Theorem 5.6, correctness is immediate from Lemmas 5.2 and 5.3. A full certifier may rebuild the entire raw union, including the observed nonflat caps, and compare it exactly with the input. The released reference code implements the flat reconstruction kernel in steps 4–6 and compares the rebuilt merged flat component; it is not an end-to-end implementation of steps 1–3 or of the full raw-union certifier.

Let N be the number of input boundary corners and k the number of returned core facets. The reference germ extractor atomises each segment against all exact endpoints and builds carrier labels during atomisation. Its conservative bound is $O(N^2 \log N)$ exact field operations and $O(N^2)$ crease atoms. The reference three-dimensional halfspace kernel enumerates triples and costs $O(k^3)$. Thus the proved reference bound is

$$O(N^2 \log N + k^3).$$

No sharper literature bound is imported into this statement.

The executable audit has two layers. E-A60 runs the intrinsic germ skeleton on every frozen six-vertex row. E-A58 runs the reconstruction/flat certification kernel. The frozen measurements are:

intrinsic-skeleton corpus rows	16,744
rows reconstructed by the intrinsic signature	16,744
rows agreeing with the historical graph on the corpus	16,744
collinear-subdivision mutations invariant	256/256
independent maximal-sheet cross-checks	256/256
flat-kernel reconstructions	16,744/16,744
sampled flat certifications	340/340
wrong answers certified	0
degenerate swallowed-corner witness	pass

The E-A60 canonical payload is

`D18C07176BAF7B729A550E9B9B8B9D9E04BB58A0C52D295243108417C9BE785A`.

The E-A58 payload in the release artifact is

`49A3FDC23406557354FEC8DDA51B4C2BB5CA9D43E11D2793C95034C0BA61FA5C`.

Finite replay is evidence about the implementation and its frozen domain. The proof of Theorem 5.6 is the intrinsic argument in the preceding sections.

8. Relation to supplied-boundary inverse problems

Ash and Bolker study recognition of supplied Dirichlet tessellations. Their Lemma 1 places a common cell boundary on the perpendicular bisector of its two sources, and their Theorem 2 constructs a Dirichlet cell by reflecting an interior point in the known facet hyperplanes [2]. The latter is exactly the forward reflection construction used in Section 2 and is cited as such.

Subsequent recognition algorithms also begin with substantially visible boundary data. Aurenhammer takes a supplied polytopal cell complex [3]; Hartvigsen takes a tessellation of Euclidean space into polyhedra and uses linear programming [10]; and Schoenberg, Ferguson, and Li recover generators from segments demarcating cell boundaries [11]. In the plane, Alonso Ferrero states the generator-recognition problem with the Voronoi edges provided [9]. Biedl, Held, and Huber reconstruct from a supplied embedded straight-line graph with rays [4]. Power-diagram detection and weighted bisector-graph reconstruction similarly start from a supplied complex, partition, or geometric graph [5, 8]. The generalized inverse Voronoi problem studied by Aloupis et al. permits added edges while retaining the original input edges [1].

Not every inverse tessellation problem receives boundary edges. Bourne, Pearce, and Roper prove that labelled Laguerre cells are uniquely determined by their volumes and centroids and recover them by convex optimization [7]. Bourne, Mulholland, Sahu, and Tant instead fit an oriented planar Voronoi diagram from ultrasonic boundary travel times in a simplified nondestructive-testing model; the geometry is unknown, their first objective can have nonunique minimizers, and the reported reconstruction is noise-sensitive [6]. These are partial- and indirect-data comparators, but their observables are respectively global labelled cell moments and travel times, not a reflected-cap raw union.

The precise distinction in the present model is that the zero-angle source layer and its facet boundaries have already been erased by the raw union. The intrinsic germ skeleton is designed to use only the boundary information that survives that erasure. This comparison is a statement about input hypotheses, not a claim that the present result is the first possible treatment of every related erased-boundary problem.

9. Limitations and open problems

The following questions are not resolved here.

- If a core vertex passes, Theorem 5.6 does not decide whether the datum remains unique or whether an exact ambiguity exists.
- The proof is three-dimensional. Higher-dimensional analogues require a new analysis of the boundary-germ strata and candidate types.
- The reference implementation assumes an exact polygonal presentation. It does not extract such a presentation from a point cloud, floating-point mesh, or arbitrary implicit-set oracle.
- E-A58 implements the flat reconstruction kernel, not the full six-step pipeline or full raw-union certifier.
- The finite corpus and deterministic mutation sample do not establish novelty, priority, or unrestricted-domain behaviour.

References

[1] Greg Aloupis, Camilo Pérez-Rosés, Guillermo Pineda-Villavicencio, Perouz Taslakian, and Pedro Trinchet-Almaguer. Fitting voronoi diagrams to planar tessellations. *arXiv preprint arXiv:1308.5550*, 2013.

[2] Peter F. Ash and Ethan D. Bolker. Recognizing dirichlet tessellations. *Geometriae Dedicata*, 19:175–206, 1985.

[3] Franz Aurenhammer. Recognising polytopal cell complexes and constructing projection polyhedra. *Journal of Symbolic Computation*, 3:249–255, 1987.

[4] Therese Biedl, Martin Held, and Stefan Huber. Recognizing straight skeletons and voronoi diagrams and reconstructing their input. In *10th International Symposium on Voronoi Diagrams in Science and Engineering*, pages 37–46, 2013.

[5] Steffen Borgwardt and Rafael M. Frongillo. Power diagram detection with applications to information elicitation. *arXiv preprint arXiv:1711.06207*, 2017.

[6] David P. Bourne, Anthony J. Mulholland, Smita Sahu, and Katherine M. M. Tant. An inverse problem for voronoi diagrams: A simplified model of non-destructive testing with ultrasonic arrays. *Mathematical Methods in the Applied Sciences*, 44(5):3727–3745, 2021.

[7] David P. Bourne, Mason Pearce, and Steven M. Roper. Inverting laguerre tessellations: Recovering tessellations from the volumes and centroids of their cells using optimal transport. *ESAIM: Mathematical Modelling and Numerical Analysis*, 59(2):841–871, 2025.

[8] Günther Eder, Martin Held, Stefan de Lorenzo, and Peter Palfrader. On the recognition and reconstruction of weighted voronoi diagrams and bisector graphs. *Computational Geometry*, 109:101935, 2023.

[9] M. Montserrat Alonso Ferrero. Voronoi diagram: The generator recognition problem. *arXiv preprint arXiv:1105.4246*, 2011.

[10] David Hartvigsen. Recognizing voronoi diagrams with linear programming. *ORSA Journal on Computing*, 4(4):369–374, 1992.

[11] Frederic Paik Schoenberg, Thomas Ferguson, and Cheng Li. Inverting dirichlet tessellations. *The Computer Journal*, 46(1):76–83, 2003.

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